Project Management Fundamentals, Planning, WBS, and Scheduling

Software Engineering & Project Management • Student Learning Book

Learning outcomes

• Explain what a software project is and why project management is needed.

• Identify the main project constraints and key stakeholders.

• Create a simple project charter and define scope.

• Break deliverables into manageable work packages using a Work Breakdown Structure (WBS).

• Estimate task duration, identify dependencies, and represent a schedule using a Gantt chart.

Running case study: A team is building a College Event Management System (CEMS) so students can browse events and register while organizers manage events and attendance.

1. What Is a Software Project?

A software project is a temporary effort undertaken to create, improve, or maintain a defined software product or service. It has an objective, a start and end, a team, constraints, and deliverables.

Project management is the application of knowledge, skills, tools, and techniques to plan and coordinate work so the team can meet agreed goals. It includes deciding what will be delivered, who will do the work, when activities should happen, how progress will be measured, and how risks or changes will be handled.

Example: A college asks a team to build CEMS before the annual festival. The team must clarify requirements, estimate effort, assign responsibilities, test the product, and deliver it in time for students to register.

2. Why Project Management Matters

• Creates a shared understanding of scope and success criteria.

• Makes work visible so dependencies and delays can be identified early.

• Helps coordinate people, tools, budget, and time.

• Makes risks and assumptions explicit.

• Supports communication with sponsors, users, developers, and testers.

• Provides a controlled way to evaluate changes and trade-offs.

A plan is not a guarantee that everything will happen exactly as expected. It is a baseline for making informed decisions when new information, risks, or changes arise.

3. Project Constraints and Trade-offs

Constraint / concern	Meaning	CEMS example
Scope	What is and is not included	Event browsing and registration are included; online ticket payment may be excluded
Time / schedule	When milestones and delivery are expected	A usable version is needed before festival registration opens
Cost / effort	Available budget and team capacity	A small student team has limited hours and hosting budget
Quality	Required behavior and acceptable product attributes	No duplicate registrations; personal data must be protected

Constraint / concern	Meaning	CEMS example
Resources	People, skills, tools, and infrastructure	One developer may also handle testing and deployment
Risk	Uncertain events that could affect goals	A key integration may not be ready on time

These concerns influence each other. Adding features may increase effort, cost, and risk. Shortening the schedule may require reducing scope, adding resources where practical, or accepting a carefully evaluated trade-off. Quality and safety should not be silently sacrificed.

4. Stakeholders and Responsibilities

A stakeholder is a person or group that can affect, be affected by, or has an interest in the project. Stakeholder identification helps the team understand needs, authority, and communication expectations.

Stakeholder	Likely interest / responsibility
College sponsor	Approves objectives, scope, budget, and major decisions
Students	Need an understandable and reliable event discovery and registration experience
Event organizers	Create events, manage capacity, and review attendees
Project manager / team lead	Coordinates scope, schedule, resources, risks, and communication
Developers	Design, implement, review, and maintain software
Testers / quality role	Plan and execute tests; report evidence and risks
IT / operations	Hosting, access, backups, deployment, and support

5. Project Charter and Scope

A project charter is a short document that authorizes a project and records its purpose, objectives, high-level scope, stakeholders, assumptions, constraints, and initial success criteria.

Sample CEMS project charter

Field	Sample content
Purpose	Provide one place for students to discover college events and register.
Objectives	Allow students to browse published events, register within capacity, and receive a confirmation; allow organizers to manage events.
In scope	Authentication, event catalogue, registration, organizer dashboard, attendance list, basic notifications
Out of scope	Online payments, external ticket marketplace, advanced analytics
Assumptions	College provides event data and identifies an organizer representative.
Constraints	Must be usable before the festival; limited team and hosting budget
Success criteria	Agreed critical workflows pass acceptance tests; event capacity and access rules work correctly

Scope should be explicit. “Build a complete event platform” is too vague to plan reliably. A clear in-scope/out-of-scope list reduces misunderstandings and helps evaluate change requests.

6. Work Breakdown Structure (WBS)

A Work Breakdown Structure decomposes the total project scope into smaller deliverables and work packages. It answers: “What work must be delivered?” A WBS is deliverable-oriented; it is not simply a list of dates or a list of people.

Example WBS for CEMS:

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1.0 College Event Management System
1.1 Project management
1.1.1 Charter and scope baseline
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1.1.2 Schedule and risk register
1.2 Requirements and design
1.2.1 Stakeholder interviews and SRS
1.2.2 UML and architecture
1.3 Implementation
1.3.1 Authentication and roles
1.3.2 Event catalogue and organizer tools
1.3.3 Registration and capacity rules
1.4 Quality assurance
1.4.1 Test plan and test cases
1.4.2 Integration and regression testing
1.5 Release and handover
1.5.1 Deployment and smoke tests
1.5.2 User guide and presentation
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Work package characteristics

• Has a clear deliverable or outcome.

• Can be estimated and assigned to a responsible person or team.

• Has completion criteria that can be checked.

• Is small enough to manage but not broken into meaningless micro-tasks.

> The 100% rule: the WBS should account for all work included in the approved project scope, including management, quality, deployment, and documentation—not just coding.

7. From WBS to Activities and Dependencies

After creating the WBS, identify the activities needed to complete each work package. Then estimate duration, identify dependencies, and assign responsibilities.

Activity	Predecessor	Illustrative duration
A. Confirm scope and requirements	None	3 days
B. Create design and UML	A	2 days
C. Implement event catalogue	B	4 days
D. Implement registration	B	5 days
E. Integrate and test	C and D	3 days
F. Fix defects and retest	E	2 days
G. Deploy and hand over	F	1 day

These durations are examples for learning, not estimates for every real project. Actual estimates should consider complexity, team experience, availability, uncertainty, and the definition of completion.

8. Gantt Charts

A Gantt chart displays activities against a calendar or timeline. Each activity is shown as a bar spanning its planned start and finish. It helps teams see overlaps, milestones, and schedule progress.

Illustrative timeline (each block represents an approximate work period):

Activity	Week 1	Week 2	Week 3	Week 4
Requirements	■■■■			
Design	■■	■■		

Activity	Week 1	Week 2	Week 3	Week 4
Implementation		■■■■	■■■■	
Testing and fixes			■■	■■
Deployment / handover				■

A Gantt chart is only as useful as its assumptions and updates. It should reflect dependencies and real progress. Overlapping tasks are safe only when the necessary inputs and people are available.

9. Estimation and Schedule Risk

Estimation is a forecast, not a promise. Teams can estimate by analogy with previous work, expert judgment, decomposition into smaller tasks, or other suitable methods. Record assumptions and uncertainty instead of presenting uncertain estimates as exact facts.

• Estimate work effort separately from elapsed calendar duration.

• Account for reviews, testing, rework, meetings, and deployment—not only implementation.

• Avoid assuming every team member is available full-time.

• Identify tasks with uncertain requirements or external dependencies.

• Update estimates when evidence changes and explain the impact on milestones.

10. Common Project Planning Mistakes

• Starting implementation before agreeing on scope and acceptance criteria.

• Creating a schedule without identifying dependencies.

• Ignoring testing, documentation, deployment, and handover.

• Making task estimates without discussing assumptions.

• Treating the first plan as fixed even when the project changes.

• Assigning responsibility vaguely so nobody owns completion.

• Adding new scope without evaluating time, cost, quality, and risk.

11. Classroom Activity: Plan CEMS

In groups, prepare a starter plan for CEMS.

1. Write a project objective and three measurable success criteria.

2. List at least five stakeholders and their needs.

3. Define in-scope and out-of-scope features.

4. Create a three-level WBS with at least 12 work packages or sub-deliverables.

5. Identify at least seven activities, their predecessors, and rough durations.

6. Draw a four-week Gantt chart and mark two milestones.

7. Identify three risks and propose one response for each.

Deliverable: a project charter, WBS, dependency table, Gantt chart, and short presentation explaining assumptions.

12. Review Questions with Suggested Answers

1. What is a software project?

Suggested answer: A temporary effort to deliver a defined software product or service with objectives, resources, constraints, and deliverables.

2. Why is project management needed?

Suggested answer: To coordinate scope, schedule, effort, quality, stakeholders, risks, and changes.

3. Name three project constraints or concerns.

Suggested answer: Scope, schedule, cost/effort, quality, resources, and risk are examples.

4. What is a project charter?

Suggested answer: A short authorization and alignment document covering purpose, objectives, scope, stakeholders, assumptions, constraints, and success criteria.

5. What does a WBS represent?

Suggested answer: A hierarchical decomposition of the complete project scope into deliverables and work packages.

6. Explain the 100% rule.

Suggested answer: The WBS should include all work in the approved scope, not just development tasks.

7. How does a WBS differ from a Gantt chart?

Suggested answer: A WBS organizes deliverables and work; a Gantt chart places activities on a timeline.

8. What is a dependency?

Suggested answer: A relationship where one activity relies on another activity or deliverable.

9. Why should testing and documentation appear in the plan?

Suggested answer: They require real effort and are part of delivering a usable, supportable product.

10. Why should estimates include assumptions?

Suggested answer: Because duration and effort depend on uncertainty, scope, resources, and working conditions.

13. Glossary

Term	Meaning
Project	Temporary effort with a defined objective and deliverables
Project management	Planning, coordinating, and controlling work toward agreed goals
Stakeholder	Person or group with an interest in or influence on the project
Project charter	Document that establishes purpose, scope, authority, and initial success criteria
Scope	Work and outcomes included in the project
WBS	Hierarchical decomposition of project scope into manageable deliverables
Work package	Manageable WBS element that can be estimated and assigned
Dependency	Relationship in which an activity relies on another
Gantt chart	Timeline visualization of activities and durations
Milestone	A significant event or checkpoint in a project schedule

14. Homework

1. Create a one-page charter for a Library Management System.
2. Prepare a three-level WBS for the system with at least 12 work packages.
3. List eight activities, estimate their duration, and identify predecessors.
4. Draw a Gantt chart showing the plan over four weeks and mark two milestones.
5. Identify three schedule risks and explain how you would respond to each.

Next chapter: Project Scheduling with PERT and CPM, Critical Path, and Schedule Estimation.